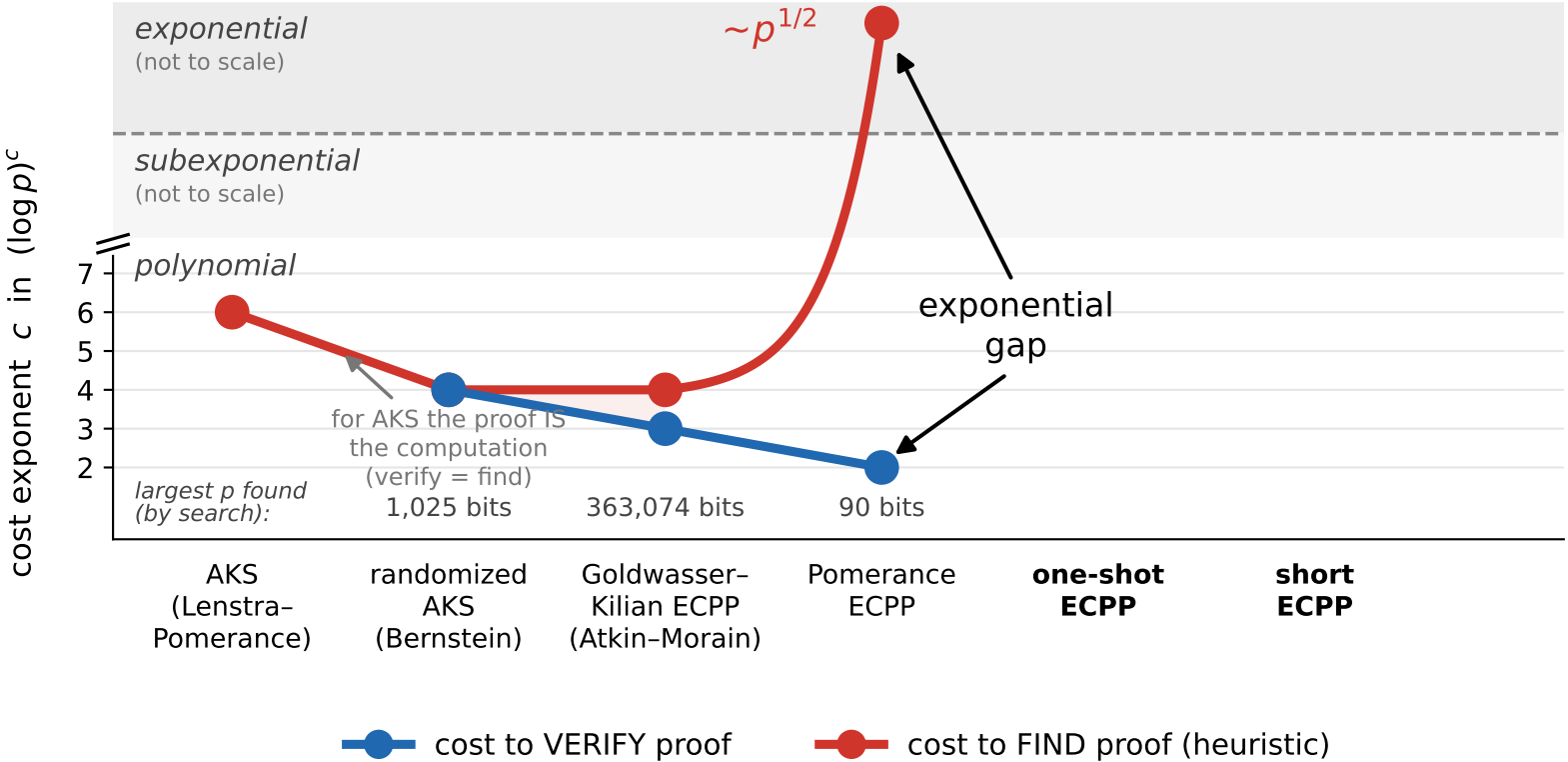
exponential
(not to scale)
$\sim p^{1/2}$
subexponential
(not to scale)
polynomial
cost exponent c in $(\log p)^c$
7
6
5
4
3
2
exponential
gap
for AKS the proof IS
the computation
(verify = find)
largest p found
(by search):
1,025 bits
363,074 bits
90 bits
AKS
(Lenstra–
Pomerance)
randomized
AKS
(Bernstein)
Goldwasser–
Kilian ECPP
(Atkin–Morain)
Pomerance
ECPP
one-shot
ECPP
short
ECPP
cost to VERIFY proof
cost to FIND proof (heuristic)